

EE5239 Introduction to Nonlinear Optimization

Lecture 8: Inequality Constraints and Duality

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Summary

- After this lecture, I hope you will be able to understand
 - KKT condition
 - Key difference between inequality and equality constraints
 - Concepts about primal/dual problems
 - The duality theorem
- **Readings**
 - Section 3.3.1, 3.4 (version 2)
 - Section 4.3.1, 4.4 (version 3)
- **Readings [advanced]**
 - The rest of Sec 3.3 (version 2)
 - The rest of Sec 4.3 (version 3)

Equality Constrained Problem

Let us first consider the following equality constrained problem

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i = 1, \dots, m.\end{array}$$

where $f : R^n \mapsto R$, $h_i : R^n \mapsto R, i = 1, \dots, m$, are continuously differentiable function.

n variables, **m** constraints

Inequality Constrained Problem

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & h(x) = 0, \quad g(x) \leq 0.\end{array}$$

where $f : R^n \mapsto R$, $h : R^n \mapsto R^m$, $g : R^n \mapsto R^r$ are continuously differentiable. Here $h = (h_1, \dots, h_m)$, $g = (g_1, \dots, g_r)$.

- **r** inequality constraints; **m** equality constraints

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- Consider the set of active inequality constraints

$$A(x) = \{j \mid g_j(x) = 0\}$$

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- r inequality constraints; m equality constraints
- Consider the set of active inequality constraints

$$A(x) = \{j \mid g_j(x) = 0\}$$

- If x^* is a local minimum:
 - The active inequality constraints at x^* can be treated as equalities
 - The inactive constraints at x^* do not matter

Inequality Constrained Problem

- Assuming regularity of x^* and assigning zero Lagrange multipliers to inactive constraints, we have

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*) + \sum_{j=1}^r u_j^* \nabla g_j(x^*) = 0, \quad u_j^* = 0, \forall j \notin A(x^*).$$

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- Extra property: $u_j^* \geq 0$ for all j . (why?)
- Intuitive reason: Relax the j -th constraint, $g_j(x) \leq q_j$, and notice

$$u_j^* = - \frac{\Delta \text{cost due to } q_j}{q_j}.$$

The above is due to the sensitivity analysis argument.

Karash-Kuhn-Tucker (KKT) Conditions

- Let x^* be a local minimum and a regular point. Then there exist unique Lagrange multiplier vectors $\lambda^* = (\lambda_1^*, \dots, \lambda_m^*)$, $u^* = (u_1^*, \dots, u_r^*)$, such that

$$\nabla_x L(x^*, \lambda^*, u^*) = 0, \quad u_j^* \geq 0, \quad j = 1, \dots, r, \quad u_j^* = 0, j \notin A(x^*). \quad (0.1)$$

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- The condition $u_j^* = 0, j \notin A(x^*)$ can be compactly written as

$$u_j^* g_j(x^*) = 0, \forall j = 1, \dots, r \quad (0.2)$$

Often referred to as the **Complementarity Condition**.

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- Don't forget that, of course, x^* also has to be feasible, that is

$$g_i(x^*) \leq 0, h_j(x^*) = 0, \quad \forall i, j. \quad (0.3)$$

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$$g_i(x^*) \leq 0, h_j(x^*) = 0, \quad \forall i, j. \quad (0.3)$$

- (1)–(3) together are called **KKT conditions**

KKT Conditions

- If f , h , and g are twice continuously differentiable,

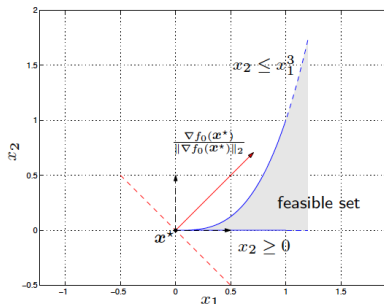
$$y' \nabla_{xx}^2 L(x^*, \lambda^*, u^*) y \geq 0, \quad \forall y \in V(x^*),$$

where

$$V(x^*) = \{y \mid \nabla h(x^*)' y = 0, \nabla g_j(x^*)' y = 0, j \in A(x^*)\}.$$

- These are called second-order necessary conditions

Example 5



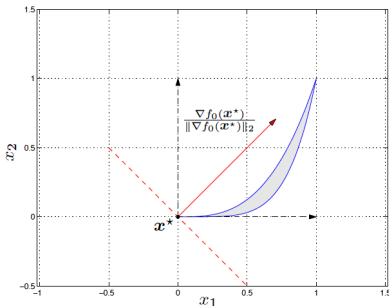
Consider the problem

$$\begin{aligned} & \underset{x \in \mathbb{R}^2}{\text{minimize}} && f_0(x) = x_1 + x_2 \\ & \text{subject to} && f_1(x) = x_2 - x_1^3 \leq 0, \\ & && f_2(x) = -x_2 \leq 0. \end{aligned}$$

This is a problem with a linear objective function $f_0(x)$, one nonlinear inequality constraint $f_1(x) \leq 0$ and one linear inequality constraint $f_2(x) \leq 0$. There are infinitely many feasible points, however, at x^* (where both inequality constraints are active), there is no λ^* satisfying

$$\nabla f_0(x^*) + \lambda_1^* \nabla f_1(x^*) + \lambda_2^* \nabla f_2(x^*) = 0.$$

Example 6



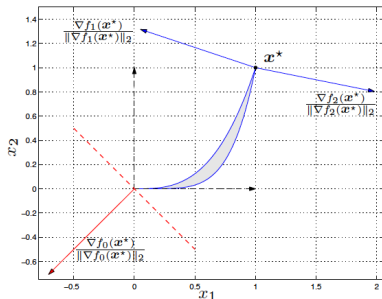
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At the solution x^* , all inequality constraints are active and their gradients are co-linear. There is no λ^* satisfying

$$\nabla f_0(x^*) + \lambda_1^* \nabla f_1(x^*) + \lambda_2^* \nabla f_2(x^*) + \lambda_3^* \nabla f_3(x^*) = 0.$$

Example 7



$$\underset{x \in \mathbb{R}^2}{\text{maximize}} \quad f_0(x) = x_1 + x_2$$

$$\text{subject to} \quad f_1(x) = x_2 - x_1^3 \leq 0,$$

$$f_2(x) = x_1^5 - x_2 \leq 0,$$

$$f_3(x) = -x_2 \leq 0.$$

Only the first two inequality constraints are active at the solution $x^* = (1, 1)$, which satisfies the KKT necessary conditions with $\lambda_1^* = 3$, $\lambda_2^* = 2$ and $\lambda_3^* = 0$. This can be verified by solving the equation

$$\begin{bmatrix} \nabla f_1(x_1) \\ \nabla f_2(x_1) \end{bmatrix}^T \begin{bmatrix} \lambda_1^* \\ \lambda_2^* \end{bmatrix} = -\nabla f(x^*) \Rightarrow \begin{bmatrix} -3 & 1 \\ 5 & -1 \end{bmatrix}^T \begin{bmatrix} \lambda_1^* \\ \lambda_2^* \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Proof of KKT Conditions

- Use equality-constraints result to obtain all the conditions except for $u_j^* \geq 0$ for $j \in A(x^*)$ (x^* is a local min in a ball of size δ)

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- Introduce the penalty functions

$g_j^+(x) = \max\{0, g_j(x)\}$, $j = 1, \dots, r$, and for $k = 1, 2, \dots$, let x^k minimize

$$f(x) + \frac{k}{2} \|h(x)\|^2 + \frac{k}{2} \sum_{j=1}^r (g_j^+(x))^2 + \frac{1}{2} \|x - x^*\|^2$$

over $\|x^k - x^*\| \leq \delta$.

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- Using the same argument as for equality constraints,

$$\lambda_i^* = \lim_{k \rightarrow \infty} k h_i(x^k), \quad i = 1, \dots, m,$$

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- Since $g_j^+(x^k) \geq 0$, we obtain $u_j^* \geq 0$ for all j .

Linear Constraints

Consider the linearly constrained problem

$$\min_{a'_j x \leq b_j, \ j=1, \dots, r} f(x).$$

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$$\min_{a'_j x \leq b_j, j=1, \dots, r} f(x).$$

- Remarkable property: No need for regularity.
- **Proposition:** If x^* is a local minimum, there exist $u^* = (u_1^*, \dots, u_r^*)$ with $u_j^* \geq 0, j = 1, \dots, r$, such that

$$\nabla f(x^*) + \sum_{j=1}^r u_j^* a_j = 0, \quad u_j^* = 0, \quad \forall j \notin A(x^*).$$

Linear Constraints

- Proof uses **Farkas Lemma**.
- Consider the cones C and $C^\perp$

$$C = \{x \mid x = \sum_{j=1}^r u_j a_j, u_j \geq 0\} \quad (0.4)$$

$$C^\perp = \{y \mid a'_j y \leq 0, j = 1, \dots, r\} \quad (0.5)$$

Then the following holds true

$$x \in C \text{ if } x'y \leq 0, \forall y \in C^\perp.$$

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- To prove, first show that C is closed (nontrivial). Then, let x be such that $x'y \leq 0, \forall y \in C^\perp$, and consider its projection $\tilde{x}$ on C . We have

$$x'(x - \tilde{x}) = \|x - \tilde{x}\|^2, \quad (x - \tilde{x})a_j \leq 0, \quad \forall j.$$

Hence, $(x - \tilde{x}) \in C^\perp$, and using the hypothesis,

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- From the above two relations, we obtain $x = \tilde{x}$, so $x \in C$.

Proof of Lagrangian Multiplier Theorem (Linear constraints)

- The local min x^* of the original problem is also a local min for the problem

$$\min_{a'_j x \leq b_j, j \in A(x^*)} f(x).$$

Hence

$$\nabla f(x^*)'(x - x^*) \geq 0, \forall x \text{ with } a'_j x \leq b_j, j \in A(x^*).$$

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- Since a constraint $a'_j x \leq b_j, j \in A(x^*)$ can also be expressed as $a'_j(x - x^*) \leq 0$, we have (let $y = (x - x^*)$)

$$\nabla f(x^*)'y \geq 0, \forall y \text{ with } a'_j y \leq 0, j \in A(x^*).$$

Proof of Lagrangian Multiplier Theorem (Linear constraints)

- From Farkas' lemma, $-\nabla f(x^*)$ has the form

$$\sum_{j \in A(x^*)} u_j^* a_j, \quad \text{for some } u_j^* \geq 0, j \in A(x^*).$$

- To complete the proof, let $u_j^* = 0$ for $j \notin A(x^*)$.

A General Sufficiency Condition

Consider the problem

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in X, \quad g_i(x) \leq 0, i = 1, \dots, r.\end{array}$$

Let x^* be feasible and μ^* satisfy

$$\mu_j^* \geq 0, \quad j = 1, \dots, r, \quad \mu_j^* = 0, \quad \forall j \notin A(x^*), \quad x^* = \operatorname{argmin}_{x \in X} L(x, \mu^*).$$

Then x^* is a global minimum of the problem.

A General Sufficiency Condition

Proof: We have

$$\begin{aligned} f(x^*) &= f(x^*) + (\mu^*)'g(x^*) = \min_{x \in X} \{f(x) + (\mu^*)'g(x)\} \\ &\leq \min_{x \in X, g(x) \leq 0} \{f(x) + (\mu^*)'g(x)\} \\ &\leq \min_{x \in X, g(x) \leq 0} f(x), \end{aligned}$$

- The first equality follows from the hypothesis, which implies that

$$(\mu^*)'g(x^*) = 0$$

- The last inequality follows from the nonnegativity of μ^*

The Dual Problem

Consider the problem

$$\min_{x \in X, a'_j x \leq b_j, j=1, \dots, r} f(x)$$

- where f is continuously differentiable over $\mathbb{R}^n$
- X is a polyhedral set (i.e., can be characterized by $Ax \leq b$)
- Note that when we talk about dual problems, we are often talking about convex problems

The Dual Problem

- Define the **dual function** $q : \mathbb{R}^r \mapsto [-\infty, \infty)$

$$q(\mu) = \inf_{x \in X} \{f(x) + \sum_{j=1}^r \mu_j (a'_j x - b_j)\}. \quad (0.6)$$

The above optimization problem is often called a **primal problem**

- Basically minimizing the Lagrangian function
- Note:** Only introduce the multipliers for a **subset of constraints**
- Define the **dual problem**

$$\max_{\mu \geq 0} \quad q(\mu). \quad (0.7)$$

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- The effective constraint set of the dual is

$$Q = \{\mu \mid \mu \geq 0, q(\mu) > -\infty\} \quad (0.8)$$

Duality Theorem

(a) If the primal problem has an optimal solution, the dual problem also has an optimal solution and the optimal values are equal.

(b) x^* is primal-optimal and μ^* is dual-optimal if and only if x^* is primal-feasible, $\mu^* \geq 0$, and

$$f(x^*) = L(x^*, \mu^*) = \min_{x \in X} L(x, \mu^*).$$

- Proof: (a) Let x^* be a primal optimal solution. For all primal feasible x , and all $\mu \geq 0$, we have $\mu'_j(a'_j x - b_j) \leq 0$ for all j , so

$$\begin{aligned} q(\mu) &\leq \inf_{x \in X, a'_j x \leq b_j, j=1, \dots, r} \left\{ f(x) + \sum_{j=1}^r \mu_j (a'_j x - b_j) \right\} \\ &\leq \inf_{x \in X, a'_j x \leq b_j, j=1, \dots, r} f(x) = f(x^*). \end{aligned}$$

Duality Theorem (Proof)

- By Lagrangian Multiplier Theorem, there exists $\mu^* \geq 0$ such that $\mu_j^*(a'_j x^* - b_j) = 0$ for all j , and $x^* = \operatorname{argmin}_{x \in X} L(x, \mu^*)$, so

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$$q(\mu^*) = L(x^*, \mu^*) = f(x^*) + \sum_{j=1}^r \mu_j^*(a'_j x^* - b_j) = f(x^*).$$

- To prove (b), suppose x^* is primal-optimal and μ^* is dual-optimal, by part (a) $f(x^*) = q(\mu^*)$, further implying

$$f(x^*) = L(x^*, \mu^*) = q(\mu^*) = \min_{x \in X} L(x, \mu^*).$$

Conversely, the relation $f(x^*) = \min_{x \in X} L(x, \mu^*)$ is written as $f(x^*) = q(\mu^*)$, and since x^* is primal-feasible and $\mu^* \geq 0$, it follows that x^* is primal optimal and μ^* is dual-optimal.

- Linear equality constraints are treated similar to inequality constraints, except that the sign of the Lagrange multipliers is unrestricted:

$$\begin{aligned} \text{Primal : } \min \quad & f(x) \\ \text{s.t.} \quad & e_i' x = d_i, i = 1, \dots, m \\ & a_j' x \leq b_j, j = 1, \dots, r \\ & x \in X. \end{aligned}$$

$$\text{Dual : } \max_{\lambda \in \mathbb{R}^m, \mu \geq 0} \inf_{x \in X} L(x, \lambda, \mu).$$